Miscellaneous Problems

1. A firm manufactures two products A and B on which the profits earned per unit are ₹ 3 and ₹ 4 respectively. Each product is processed on two machines M_1 and M_2. Product A requires one minute of processing time on M_1 and two minutes on M_2, While B requires one minute on M_1 and one minute on M_2. Machine M_1 is available for not more than 7 hrs 30 minutes while M_2 is available for 10 hrs during any working day. Formulate this problem as a linear programming problem to maximize the profit.

2. A firm manufactures pills in two sizes A and B. Size A contains 2 mgs of aspirin, 5 mgs of bicarbonate and 1 mg of codeine. Size B contains 1 mg. of aspirin, 8 mgs. of bicarbonate and 6 mgs. of codeine. It is found by users that it requires atleast 12 mgs. of aspirin, 74 mgs.of bicarbonate and 24 mgs. of codeine for providing immediate relief. It is required to determine the least number of pills a patient should take to get immediate relief. Formulate the problem as a standard LLP.

3. Solve the following linear programming problem graphically.

   Maximise $Z = 4x_1 + x_2$ subject to the constraints $x_1 + x_2 \leq 50$; $3x_1 + x_2 \leq 90$ and $x_1 \geq 0$, $x_2 \geq 0$.

4. Solve the following linear programming problem graphically.

   Minimize $Z = 200x_1 + 500x_2$ subject to the constraints: $x_1 + 2x_2 \geq 10$; $3x_1 + 4x_2 \leq 24$ and $x_1 \geq 0, x_2 \geq 0$.

5. Solve the following linear programming problem graphically.

   Maximize $Z = 3x_1 + 5x_2$ subject to the constraints: $x_1 + x_2 \leq 6$, $x_1 \leq 4$; $x_2 \leq 5$, and $x_1, x_2 \geq 0$.

6. Solve the following linear programming problem graphically. Maximize $Z = 60x_1 + 15x_2$ subject to the constraints: $x_1 + x_2 \leq 50$; $3x_1 + x_2 \leq 90$ and $x_1, x_2 \geq 0$.

7. Draw a network diagram for the following activities.

Activity code	A	B	C	D	E	F	G	H	I	J	K
Predecessor activity	–	A	A	A	B	C	C	C,D	E,F	G,H	I,J

8. Draw the network diagram for the following activities

Activity code	A	B	C	D	E	F	G
Predecessor activity	–	–	A	A	B	C	D,E

9. A Project has the following time schedule

Activity	1-2	2-3	2-4	3-5	4-6	5-6
Duration (in days)	6	8	4	9	2	7

   Draw the network for the project, calculate the earliest start time, earliest finish time, latest start time and latest finish time of each activity and find the critical path. Compute the project duration.

10. The following table gives the characteristics of project

Activity	1-2	1-3	2-3	3-4	3-5	4-6	5-6	6-7
Duration (in days)	5	10	3	4	6	6	5	5

    Draw the network for the project ,calculate the earliest start time, earliest finish time, latest start time and latest finish time of each activity and find the critical path. Compute the project duration.

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